$^{9}Be(^{37}Ca,^{34}Cl\gamma)$ 2010Do03

$^{37}Ca \rightarrow 3p + ^{34}Cl$ on ^{9}Be target.

2010Do03: a 420-MeV/nucleon ^{40}Ca primary beam was provided from the heavy ion synchrotron SIS at GSI and impinged on a 4-mg/cm^2 ^{9}Be target. ^{37}Ca was selected by FRS and was incident on a 700-mg/cm^2 secondary ^{9}Be target at 195.7 MeV/u. ^{34}Cl was produced via a secondary three proton removal reaction. γ rays were detected using the RISING setup consisting of 15 Cluster HPGe detectors, 8 MINIBALL HPGe detectors, and 8 Hector BaF_2 detectors. Fragments were detected using the calorimeter telescope CATE consisting of position sensitive Si ΔE detectors and CsI(Tl) E_{res} detectors. Measured Eγ, Iγ. Deduced lifetime using Doppler Shift Attenuation lineshape analysis (2010Do03) and peak-shift analysis (2007DoZV).

^{34}Cl Levels

E(level)†	$T_{1/2}$	Comments
0		
461	4.0 ps *9*	$T_{1/2}$: original $T_{1/2}$=4.0 ps *9* (stat) *+13−8* (syst) from lineshape analysis with Cluster detectors (2010Do03). $T_{1/2}$=2.2 ps *20* (stat) *36* (syst) from lineshape analysis with MINIBALL detectors (2010Do03). $T_{1/2}$=5.0 ps *+20−15* from peak-shift analysis with Cluster detectors (2007DoZV). $T_{1/2}$=6 ps *2* from peak-shift analysis with MINIBALL detectors (2007DoZV).
666	13 ps *5*	$T_{1/2}$: from peak-shift analysis with the MINIBALL detectors (2007DoZV). Other: 32 ps *+10−14* from peak shift analysis with the Cluster detectors (2007DoZV).
1230		

† Rounded from the Adopted Levels.

$\gamma(^{34}Cl)$

E_γ†	I_γ‡	E_i(level)	E_f
461	100	461	0
666		666	0
769	12 *6*	1230	461

† Rounded from the Adopted Gammas.
‡ From 2010Do03.

^{9}Be(^{37}Ca,^{34}Clγ) 2010Do03

Level Scheme

Intensities: Relative I_γ